Linear Regression

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lm() function

Y = x^2 example

Define some variables

```
v40 <- c(1:10)
v41 <- v40**2
```

Fit a model to predict v41 with v40

Store our linear model output

```
fit <- lm(v41 ~ v40)
fit
```

```
##
## Call:
## lm(formula = v41 ~ v40)
##
## Coefficients:
## (Intercept)          v40
##         -22           11
```

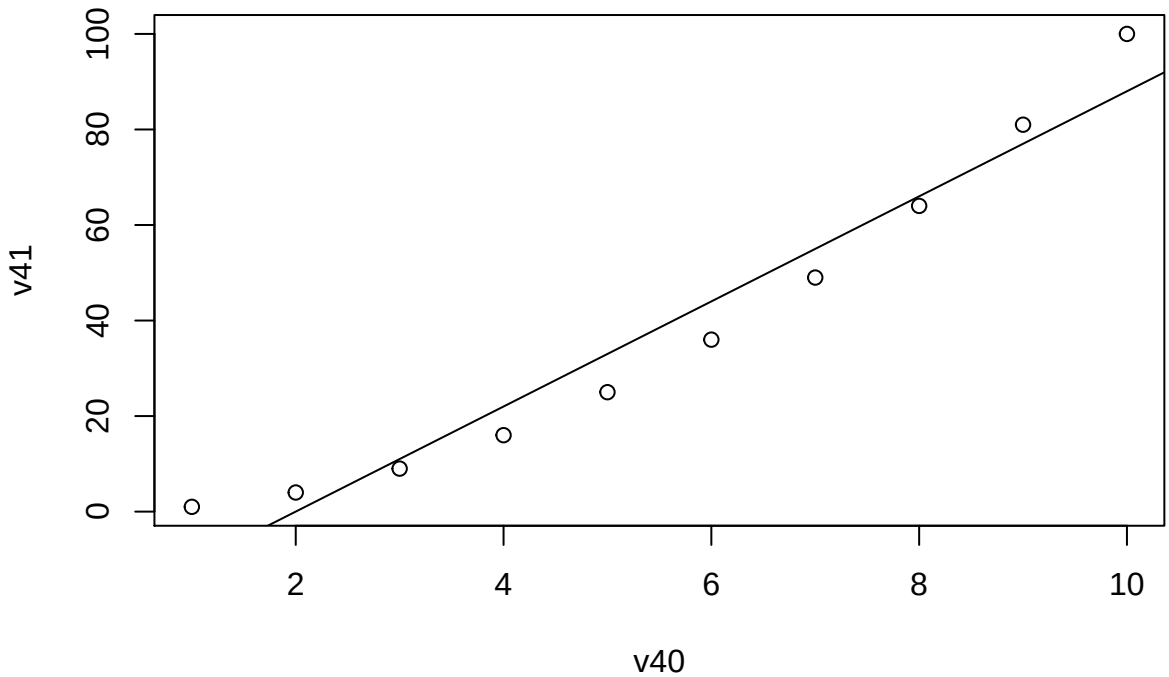
Special case of summary() function on fit

```
summary(fit)
```

```
##
## Call:
## lm(formula = v41 ~ v40)
##
## Residuals:
##    Min     1Q Median     3Q    Max
##     -8     -6     -2      4     12
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -22.0000     5.5498  -3.964  0.00415 **
## v40          11.0000     0.8944  12.298 1.78e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.124 on 8 degrees of freedom
## Multiple R-squared:  0.9498, Adjusted R-squared:  0.9435
## F-statistic: 151.3 on 1 and 8 DF,  p-value: 1.778e-06
```

See what our model looks like

```
plot(v40, v41)
abline(fit)
```



Plotting lm models

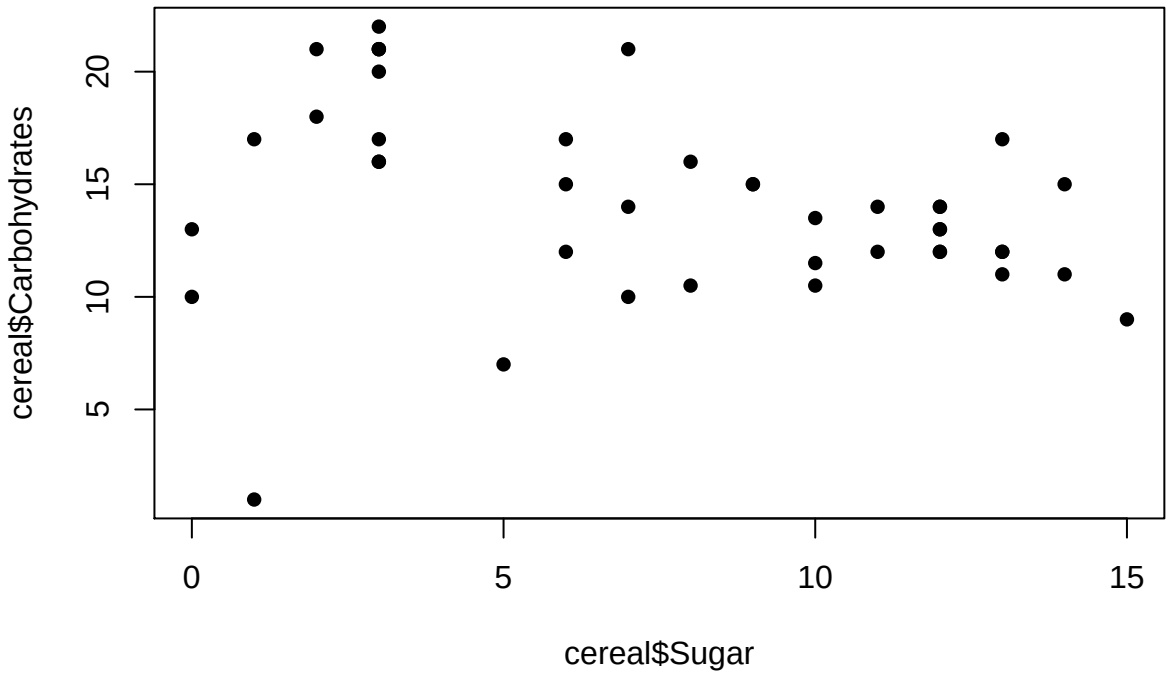
```
cereal <- read.csv('cereal.csv')
names(cereal)[1] <- 'Cereal'
head(cereal)
```

```
##             Cereal Calories Protein Fat Sodium Fiber Carbohydrates Sugar
## 1       ACCheerios      110       2   2    180   1.5          10.5    10
## 2         Cheerios      110       6   2    290   2.0          17.0     1
## 3       CocoaPuffs      110       1   1    180   0.0          12.0    13
## 4     CountChocula      110       1   1    180   0.0          12.0    13
## 5    GoldenGrahams      110       1   1    280   0.0          15.0     9
## 6 HoneyNutCheerios      110       3   1    250   1.5          11.5    10
##   Potassium
## 1        70
## 2       105
## 3        55
## 4        65
## 5        45
## 6        90
```

```r
View(cereal)
```

Let's review basic plotting, lm() function use, and abline() function use with different columns

```
plot(cereal$Sugar, cereal$Carbohydrates, pch= 16)
```



```r
fit_carb_sugar <- lm(cereal$Carbohydrates ~ cereal$Sugar)
fit_carb_sugar
```

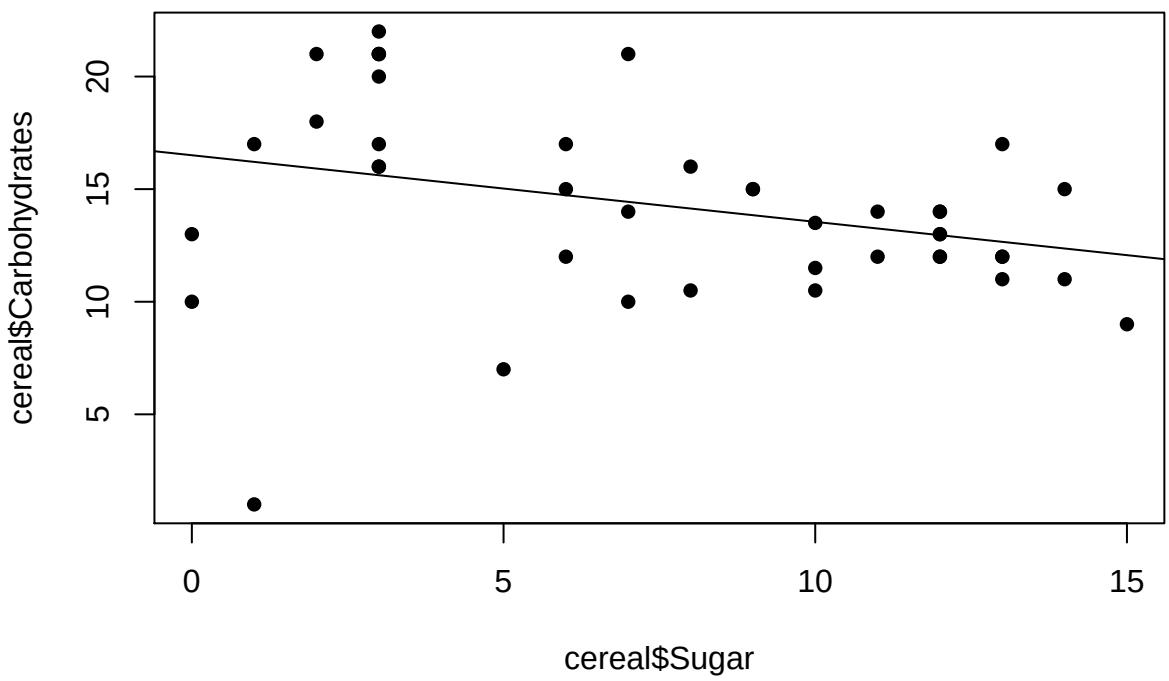
```
##
## Call:
## lm(formula = cereal$Carbohydrates ~ cereal$Sugar)
##
## Coefficients:
##  (Intercept)  cereal$Sugar
##      16.5052       -0.2958
```

```r
summary(fit_carb_sugar)
```

```
##
## Call:
## lm(formula = cereal$Carbohydrates ~ cereal$Sugar)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.2095  -1.5120   0.2695   1.9737   6.5653
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   16.5052     1.2279  13.442   <2e-16 ***
## cereal$Sugar  -0.2958     0.1391  -2.127   0.0395 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.089 on 41 degrees of freedom
## Multiple R-squared:  0.09935,    Adjusted R-squared:  0.07738
## F-statistic: 4.522 on 1 and 41 DF,  p-value: 0.03952
```

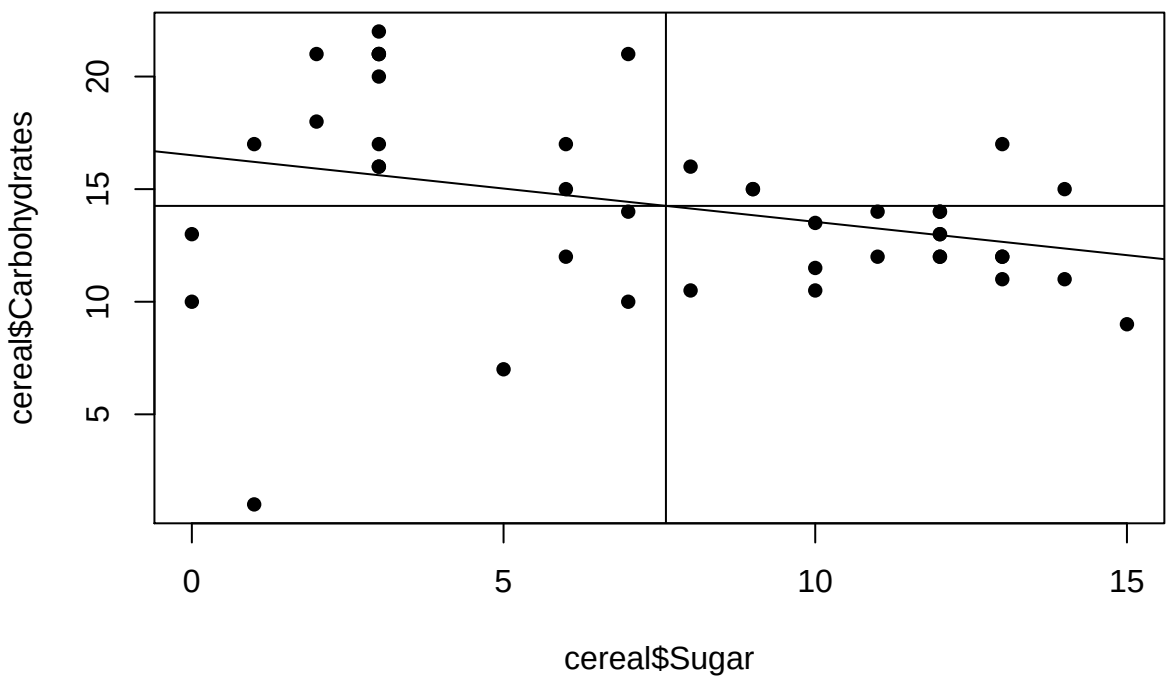
Add a line to scatterplot corresponding to the fit_carb_sugar model, a vertical line for the mean of Sugar, and a horizontal line for the mean of Carbohydrates.

```
plot(cereal$Sugar, cereal$Carbohydrates, pch= 16)
abline(fit_carb_sugar)
```

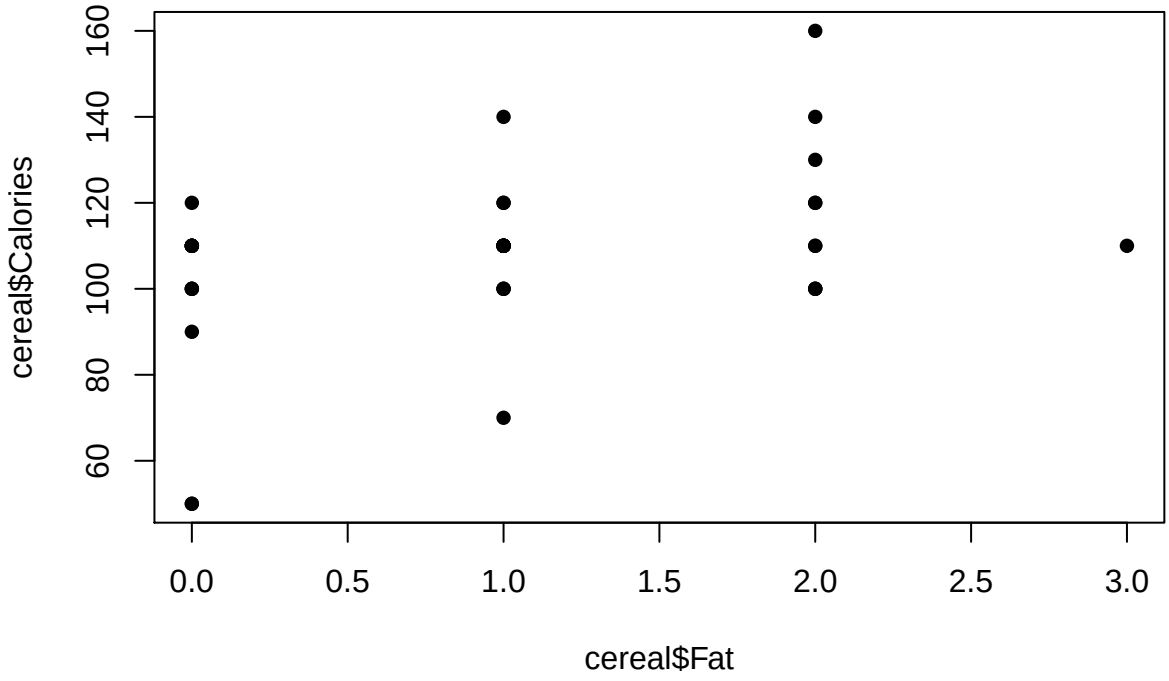


Note: the line corresponding to the model should run though the intersection of the vertical and horizontal lines for the means.

```r
plot(cereal$Sugar, cereal$Carbohydrates, pch= 16)
abline(fit_carb_sugar)
abline(h = mean(cereal$Carbohydrates))
abline(v = mean(cereal$Sugar))
```



```r
plot(cereal$Fat, cereal$Calories, pch = 16)
```



```r
fit_fat_calories <- lm(cereal$Calories ~ cereal$Fat)
fit_fat_calories
```

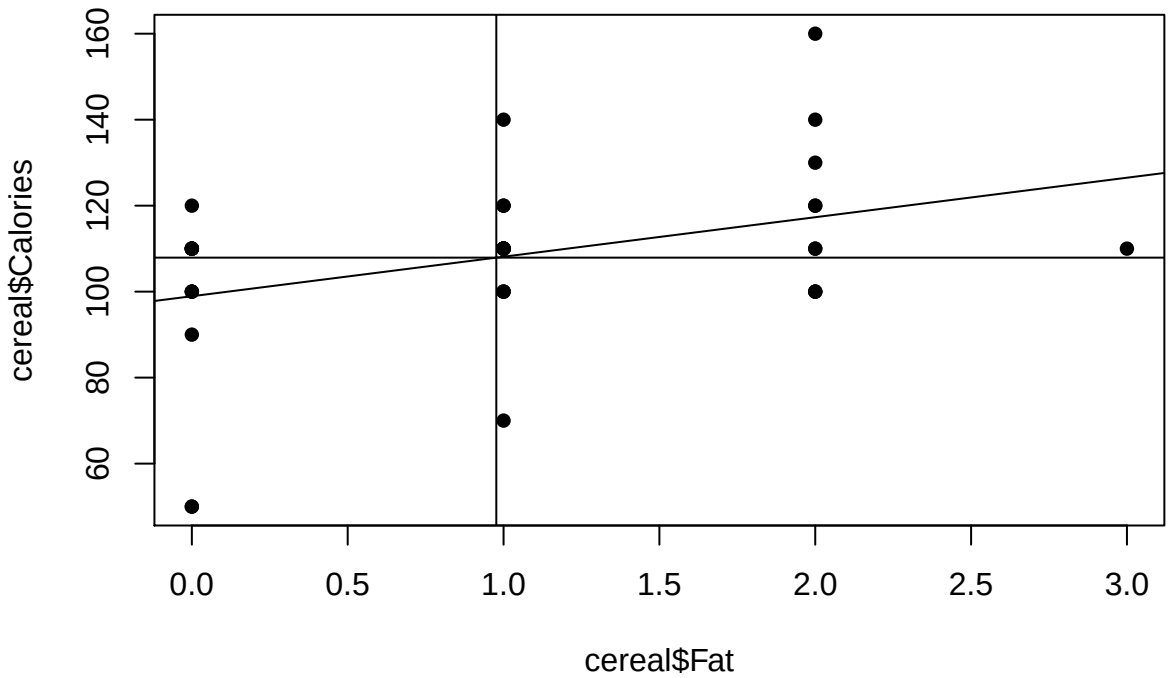
```
##
## Call:
## lm(formula = cereal$Calories ~ cereal$Fat)
##
## Coefficients:
## (Intercept)   cereal$Fat
##       98.93         9.19
```

```r
summary(fit_fat_calories)
```

```
##
## Call:
## lm(formula = cereal$Calories ~ cereal$Fat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -48.931  -7.716   1.879  11.069  42.690
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   98.931      4.284  23.095   <2e-16 ***
## cereal$Fat     9.190      3.406   2.698   0.0101 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 17.69 on 41 degrees of freedom
## Multiple R-squared:  0.1508, Adjusted R-squared:   0.13
## F-statistic: 7.278 on 1 and 41 DF,  p-value: 0.01009
```

Add line for model and lines for the means like above

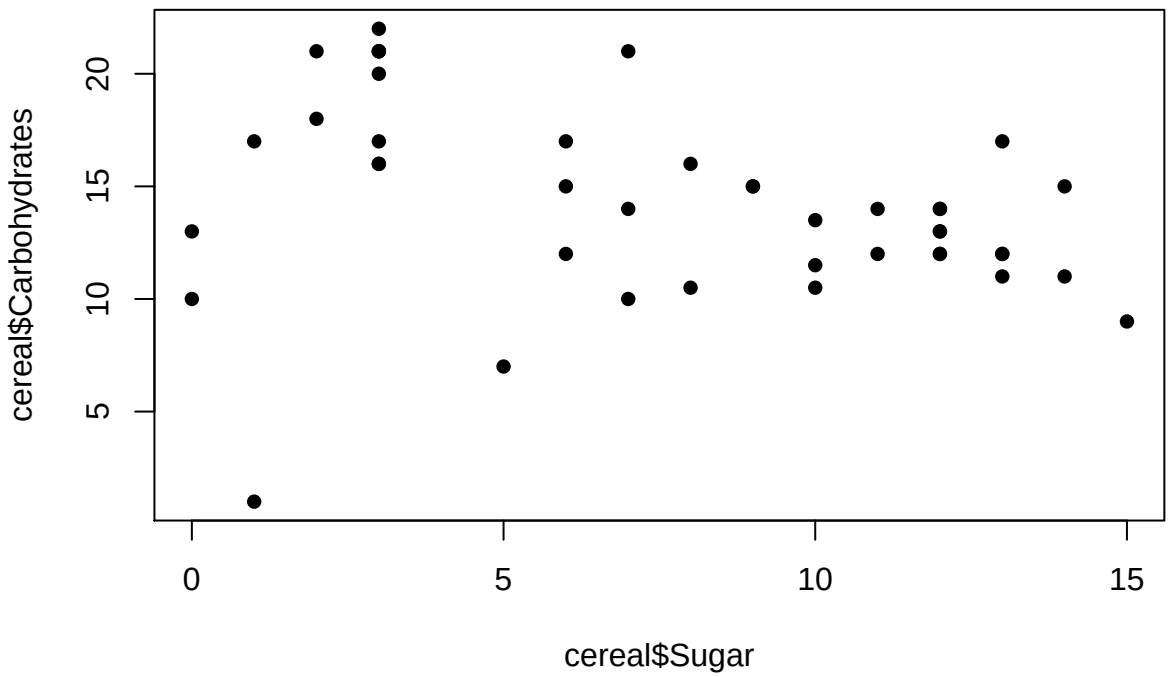
```
plot(cereal$Fat, cereal$Calories, pch= 16)
abline(fit_fat_calories)
abline(h = mean(cereal$Calories))
abline(v = mean(cereal$Fat))
```



Remove intercept

Let's make a linear model with intercept = 0

```
plot(cereal$Sugar, cereal$Carbohydrates, pch= 16)
```



```r
fit_no_intercept <- lm(cereal$Carbohydrates~ cereal$Sugar + 0)
fit_no_intercept
```

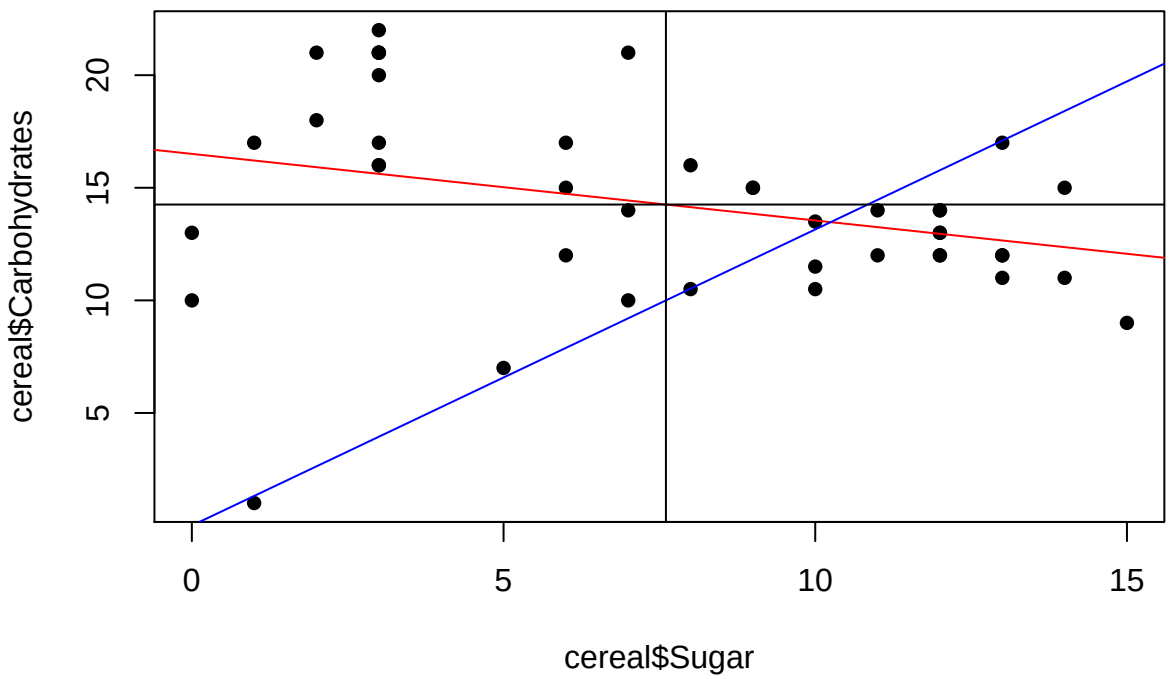
```
##
## Call:
## lm(formula = cereal$Carbohydrates ~ cereal$Sugar + 0)
##
## Coefficients:
## cereal$Sugar
##        1.315
```

```r
summary(fit_no_intercept)
```

```
##
## Call:
## lm(formula = cereal$Carbohydrates ~ cereal$Sugar + 0)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.7225  -2.5557   0.7962  12.0555  18.3703
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## cereal$Sugar   1.3148     0.1623   8.102 4.04e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.395 on 42 degrees of freedom
## Multiple R-squared:  0.6098, Adjusted R-squared:  0.6005
## F-statistic: 65.64 on 1 and 42 DF,  p-value: 4.036e-10
```

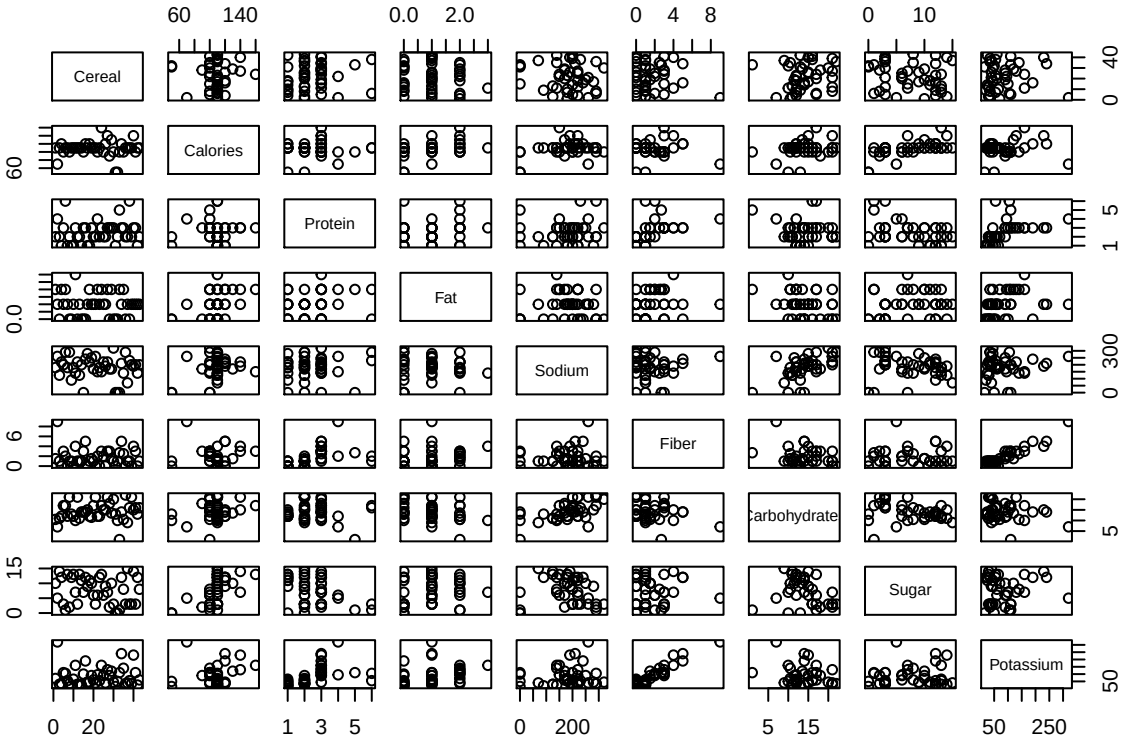
Add lines for the means as above, a red line corresponding to the default fitted model, and a blue line corresponding to the no-intercept model.

```
plot(cereal$Sugar, cereal$Carbohydrates, pch= 16)
abline(fit_carb_sugar, col = 'red')
abline(fit_no_intercept, col = 'blue')
abline(h = mean(cereal$Carbohydrates))
abline(v = mean(cereal$Sugar))
```



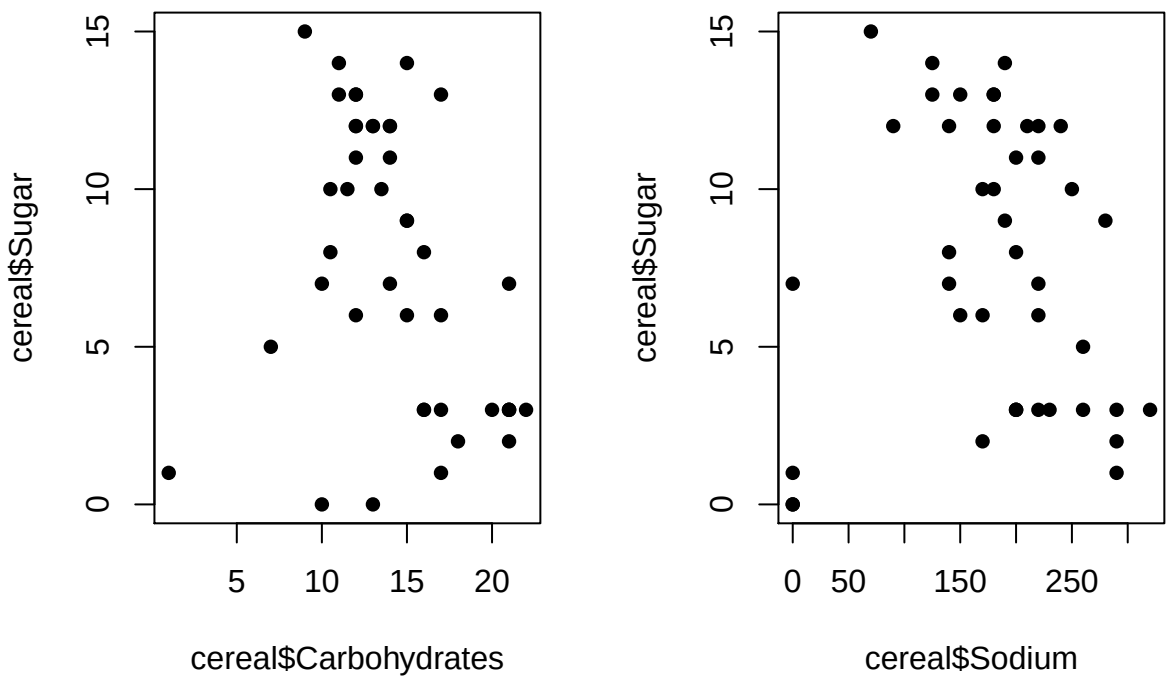
Create a model with 2 predictor (X) variables

```r
pairs(cereal)
```



Looking at the above scatterplots, I chose to predict Sugar with Carbohydrates and Sodium

```
par(mfrow = c(1,2))
plot(cereal$Carbohydrates, cereal$Sugar, pch = 16)
plot(cereal$Sodium, cereal$Sugar, pch = 16)
```



```r
fit_sugar_multiple <- lm(cereal$Sugar ~ cereal$Carbohydrates + cereal$Sodium)
fit_sugar_multiple
```

```
##
## Call:
## lm(formula = cereal$Sugar ~ cereal$Carbohydrates + cereal$Sodium)
##
## Coefficients:
##          (Intercept)  cereal$Carbohydrates         cereal$Sodium
##             12.06352              -0.45171               0.01097
```

```r
summary(fit_sugar_multiple)
```

```
##
## Call:
## lm(formula = cereal$Sugar ~ cereal$Carbohydrates + cereal$Sodium)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.6118  -2.9862   0.6391   3.5039   6.9693
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          12.06352    2.36374   5.104 8.51e-06 ***
## cereal$Carbohydrates -0.45171    0.19146  -2.359   0.0233 *
## cereal$Sodium         0.01097    0.01029   1.067   0.2925
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.35 on 40 degrees of freedom
## Multiple R-squared:  0.1243, Adjusted R-squared:  0.08047
## F-statistic: 2.838 on 2 and 40 DF,  p-value: 0.0704
```